

Plug Puller Dial Quick Start

The dials of the four Customizer steps of both pullers, one per page: which file to open, then what each step's dial moves.

Model version 0.12.0

Which file to open

Measure your plug's thickness. Up to 24 mm: open the one-sided puller, `src/Plug_Puller_Parametric.scad`. Thicker than 24 mm, or a plug you want held from both sides such as a USB-C tip or a round extension-cord plug: open the two-sided puller, `src/Plug_Puller_Two_Sided.scad`.

Each file has four steps at the top of its Customizer: your plug, your size, the attachment, and one last choice (the hook hand in the one-sided file; the print layout in the two-sided file). The dials on the next pages are those steps and nothing else.

If red text appears beside the part in the preview, read it: it names the measurement to fix, and the part will not fit until it is gone.

In every picture: black = the tool at its defaults; teal = the plug you measured; red dashed = what this dial moves.

The dial names are written exactly as the Customizer shows them. Every dial of both files is in the full dial reference.

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Two-sided puller

Step 1 - Your Plug

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Step 2 - Size

- `size`: Hand size
- `measure_finger_width`: Finger width

Step 3 - Attachment

- `attachment`: Attachment
- `strap_width`: Strap width

Step 4 - Print Layout

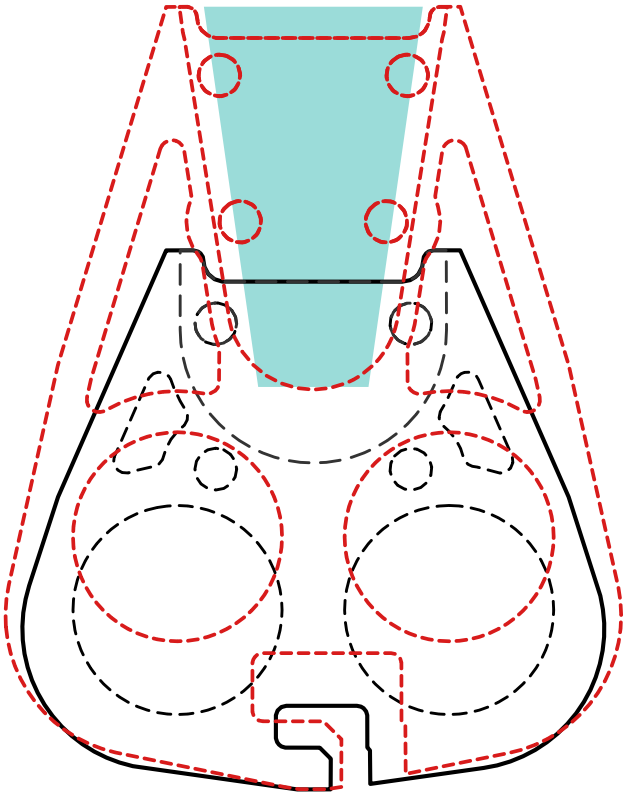
- `print_layout`: Print layout

plug_preset

One-sided puller · Step 1 - Your Plug

Plug preset

plug_preset
Measure my plug → Standard 3-prong plug - NEMA 5-15



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Fills in all six plug numbers from a measured reference plug, so the pocket, the wall notch, the cord hook and the zip-tie and wing positions all take that plug's shape at once.

Default	Range	Step	Unit
Measure my plug	5 options	—	—

Options: Measure my plug, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15, Heavy-duty extension cord - NEMA 5-15, Wide 2-prong appliance plug - NEMA 1-15.

Before: Measure my plug. After: Standard 3-prong plug - NEMA 5-15.

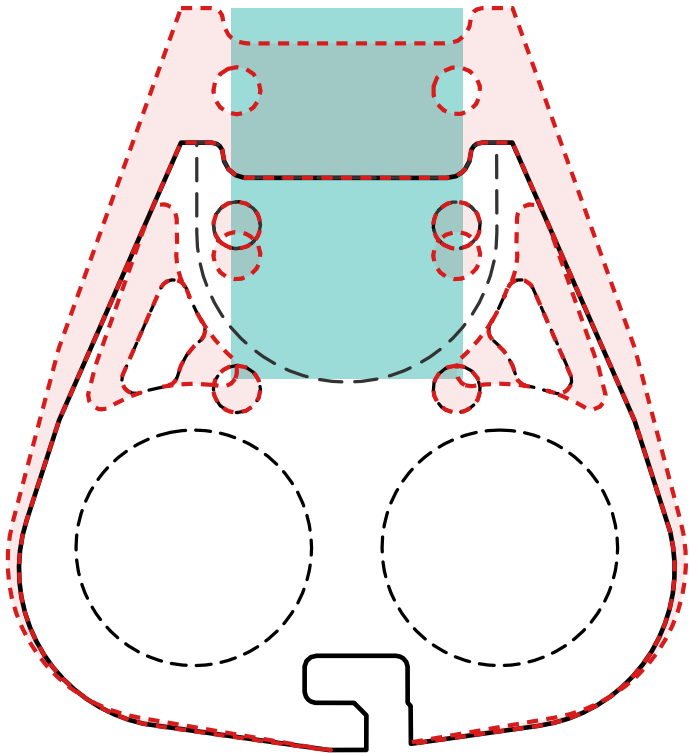
Moves: body edge, pocket, seat, wall notch, wing openings, zip-tie holes.

measure_plug_length

One-sided puller · Step 1 - Your Plug

Plug length

measure_plug_length
25.5 → 40



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Runs the pocket farther toward the finger holes and stretches the whole body to keep the finger holes and zip-tie rows clear of it.

Default	Range	Step	Unit
25.5	12 to 85	0.5	mm

Before: 25.5. After: 40.

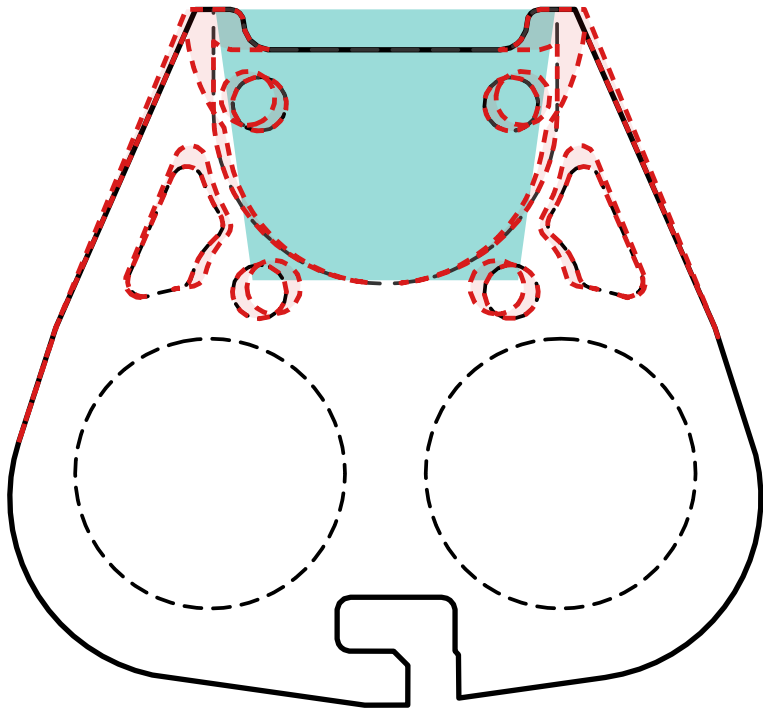
Moves: body edge, seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_prong_end

One-sided puller · Step 1 - Your Plug

Plug width at the prong end

measure_plug_width_prong_end
25 → 32



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens or narrows the pocket, its seat and the wall notch at the plug end, and moves the zip-tie holes and wing openings with them.

Default	Range	Step	Unit
25	8 to 38	0.5	mm

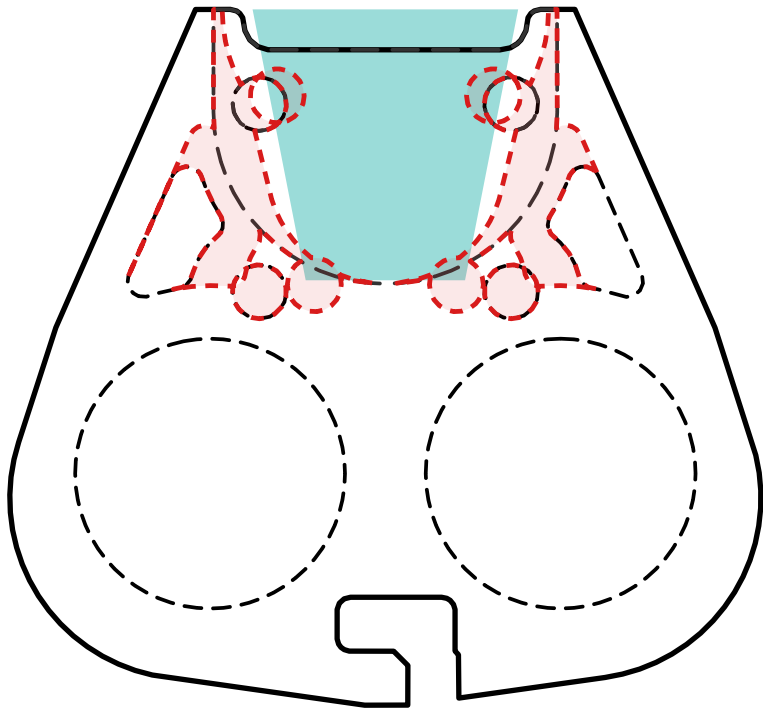
Before: 25. After: 32.

Moves: seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_cord_end

One-sided puller · Step 1 - Your Plug
Plug width at the cord end

measure_plug_width_cord_end
25 → 15



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Tilts the pocket's side walls to match the plug's taper, and the zip-tie holes and wing openings lean in with them.

Default	Range	Step	Unit
25	8 to 38	0.5	mm

Before: 25. After: 15.

Moves: pocket, wing openings.

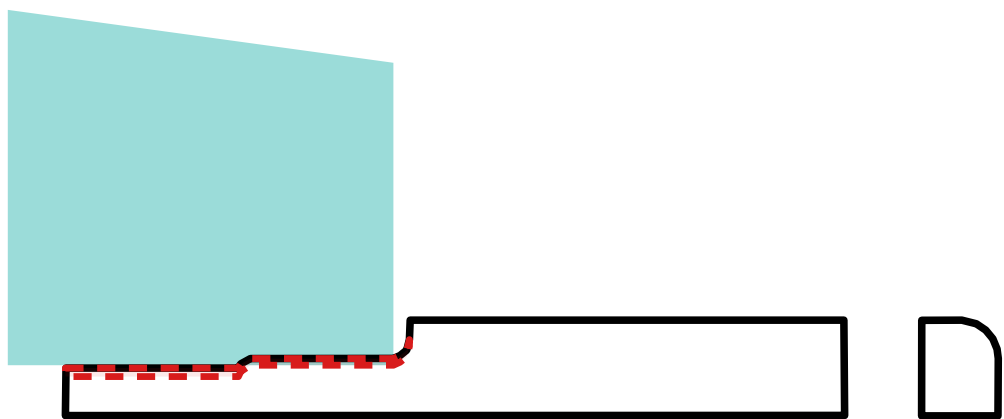
measure_plug_thickness_prong_end

One-sided puller · Step 1 - Your Plug

Plug thickness at the prong end

measure_plug_thickness_prong_end

20 → 23.5
(section-x: 0)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Deepens or shallows the two pocket floors, the round seat and the plug recess, to suit the fatter of the two thickness numbers.

Default	Range	Step	Unit
20	4 to 40	0.5	mm

Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

Moves: pocket, seat.

measure_plug_thickness_cord_end

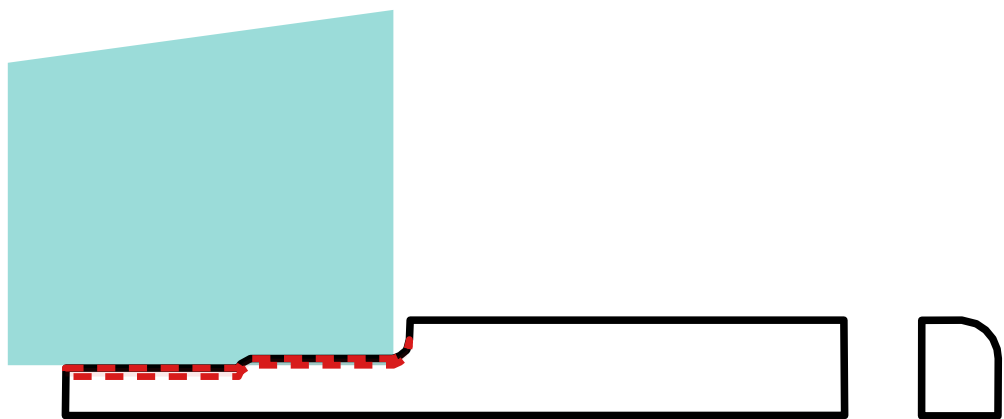
One-sided puller · Step 1 - Your Plug

Plug thickness at the cord end

measure_plug_thickness_cord_end

20 → 23.5

(section-x: 0)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Deepens or shallows the two pocket floors, the round seat and the plug recess, when this end is the fatter end of the plug.

Default	Range	Step	Unit
20	4 to 40	0.5	mm

Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

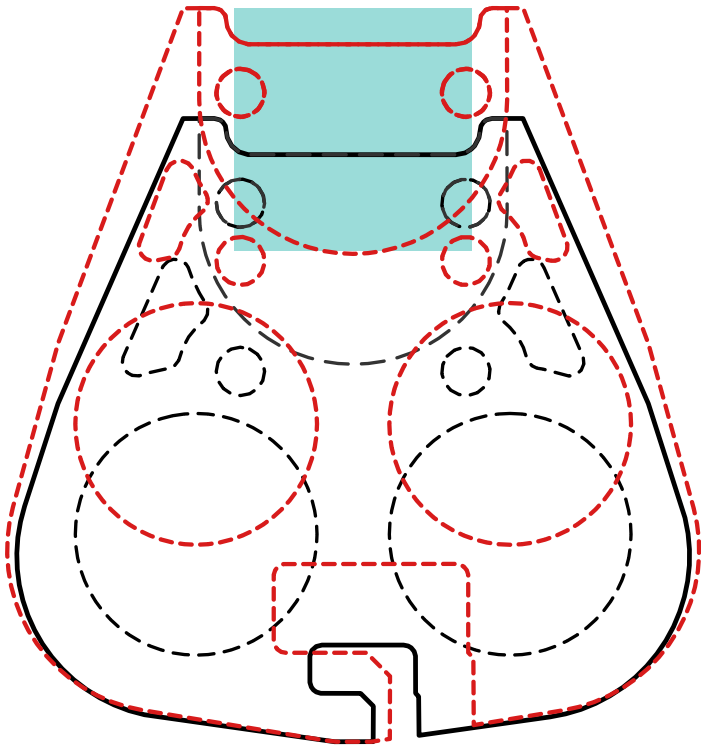
Moves: pocket, seat.

measure_cord_thickness

One-sided puller · Step 1 - Your Plug

Cord thickness

measure_cord_thickness
4 → 8



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the cord hook's slot and crossbar to fit the cord, and moves the finger holes and the whole body up to make room.

Default	Range	Step	Unit
4	1.5 to 9	0.5	mm

Before: 4. After: 8.

Moves: body edge, finger holes, pocket, wall notch, wing openings, zip-tie holes.

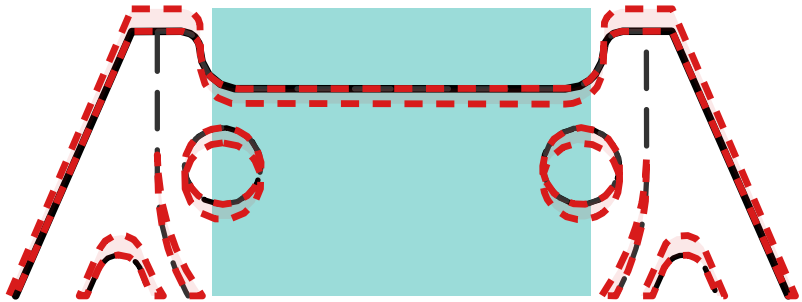
measure_wall_plate_style

One-sided puller · Step 1 - Your Plug

Outlet cover plate style

measure_wall_plate_style

Standard flat plate → Oversized / Jumbo



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Cuts the wall notch at the plug end deeper or shallower to suit the cover plate, and the zip-tie rows shift down with it.

Default	Range	Step	Unit
Standard flat plate	4 options	—	—

Options: Standard flat plate, Rocker / Decora, Oversized / Jumbo, No plate / flush.

Before: Standard flat plate. After: Oversized / Jumbo.

Moves: body edge, pocket, wall notch, wing openings, zip-tie holes.

show_plug_preview

One-sided puller · Step 1 - Your Plug

Show the see-through plug

show_plug_preview

This dial changes no shape.

Changes no shape: the see-through plug is a preview aid and is never exported.

Shows or hides the see-through plug in the preview and changes nothing in the printed tool.

Default	Range	Step	Unit
true	true / false	—	—

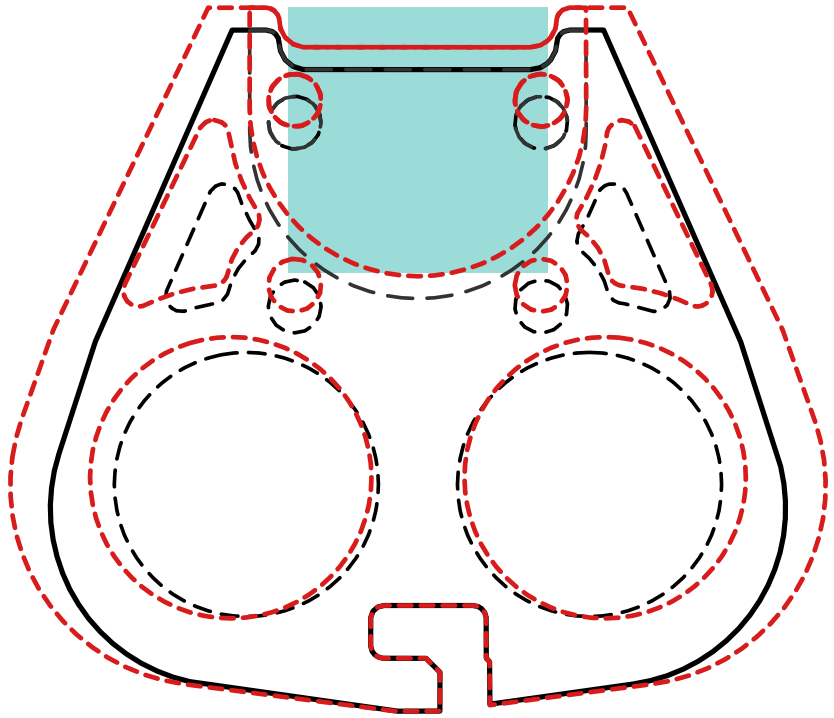
This dial changes no shape. Changes no shape: the see-through plug is a preview aid and is never exported.

size

One-sided puller · Step 2 - Size

Hand size

size
Medium → Large



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Scales the whole body, both finger holes, the wing openings and the zip-tie grid to the chosen hand size.

Default	Range	Step	Unit
Medium	5 options	—	—

Options: Small, Medium, Large, Measure my hand, Custom.

Before: Medium. After: Large.

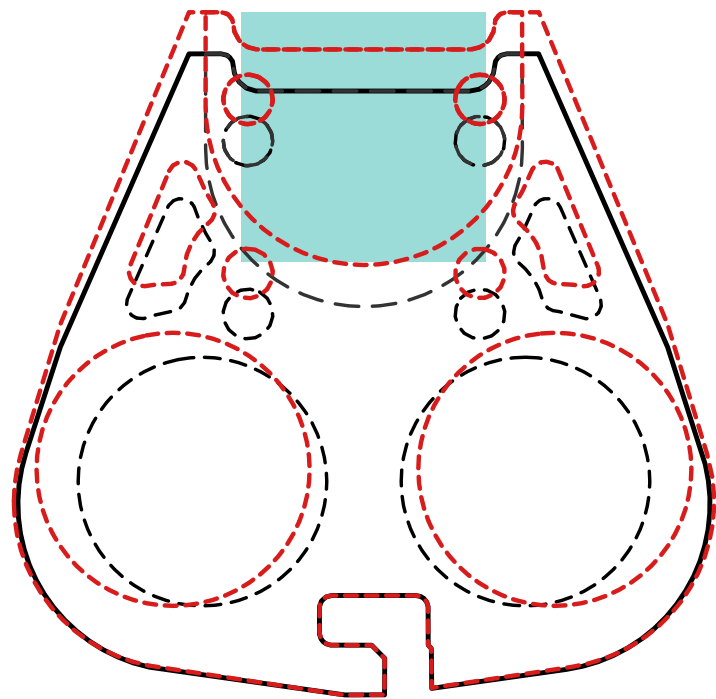
Moves: body edge, finger holes, pocket, wing openings, zip-tie holes.

measure_finger_width

One-sided puller · Step 2 - Size

Finger width

measure_finger_width
20 → 26
(size: Measure my hand)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes, spaces them farther apart and pushes them and the body up to keep the walls printable.

Default	Range	Step	Unit
20	14 to 32	0.5	mm

Before: 20. After: 26. Context: size = Measure my hand.

Only acts when size is Measure my hand.

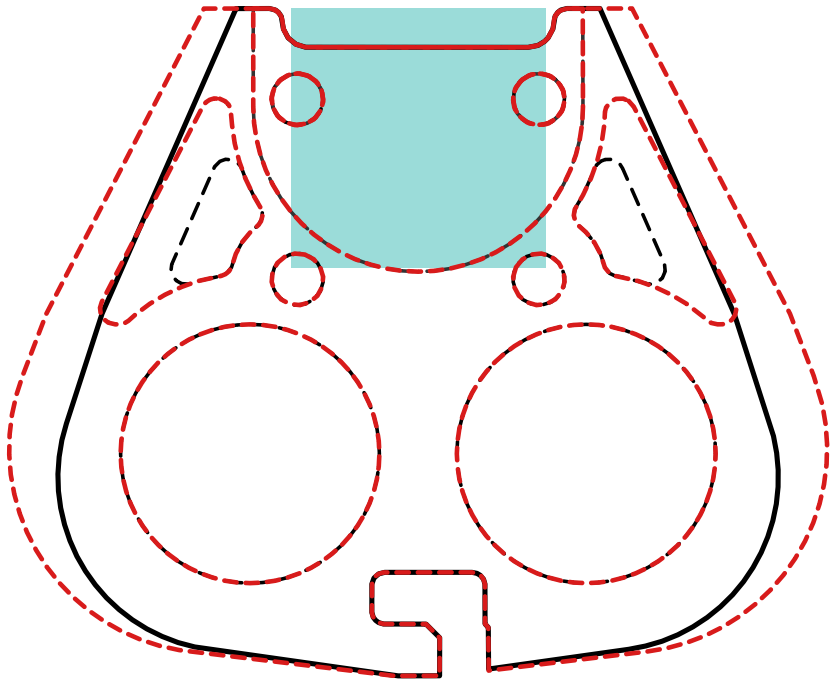
Moves: finger holes, pocket, wall notch, wing openings.

measure_hand_width

One-sided puller · Step 2 - Size

Hand width

measure_hand_width
85 → 100
(size: Measure my hand)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the whole body outline and thickens the slab in step with the hand.

Default	Range	Step	Unit
85	60 to 110	1	mm

Before: 85. After: 100. Context: size = Measure my hand.

Only acts when size is Measure my hand.

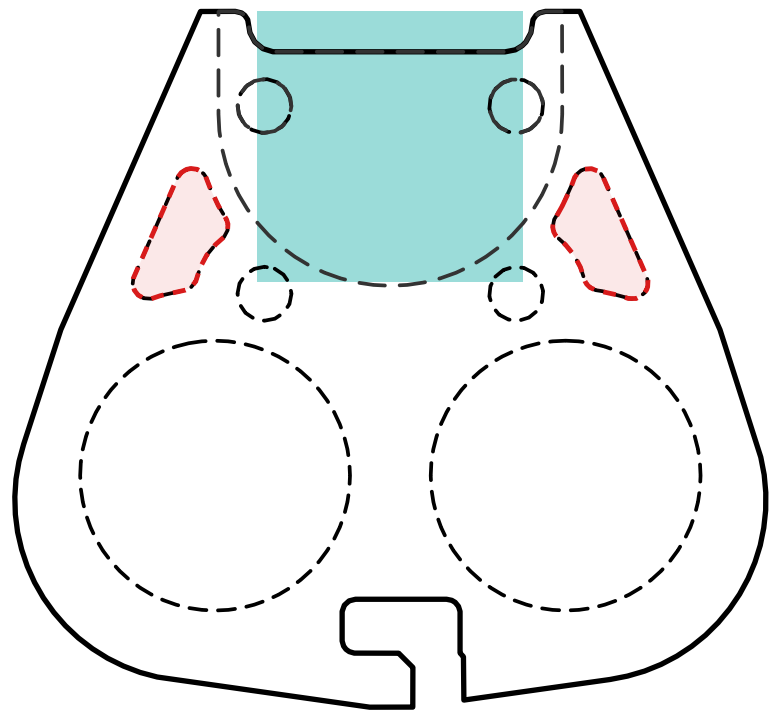
Moves: wing openings.

attachment

One-sided puller · Step 3 - Attachment

Attachment

attachment
Zip ties + Velcro → Zip ties



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Adds or removes the zip-tie holes and the wing or slot openings for the strap.

Default	Range	Step	Unit
Zip ties + Velcro	4 options	—	—

Options: Zip ties, Velcro strap, Zip ties + Velcro, None.

Before: Zip ties + Velcro. After: Zip ties.

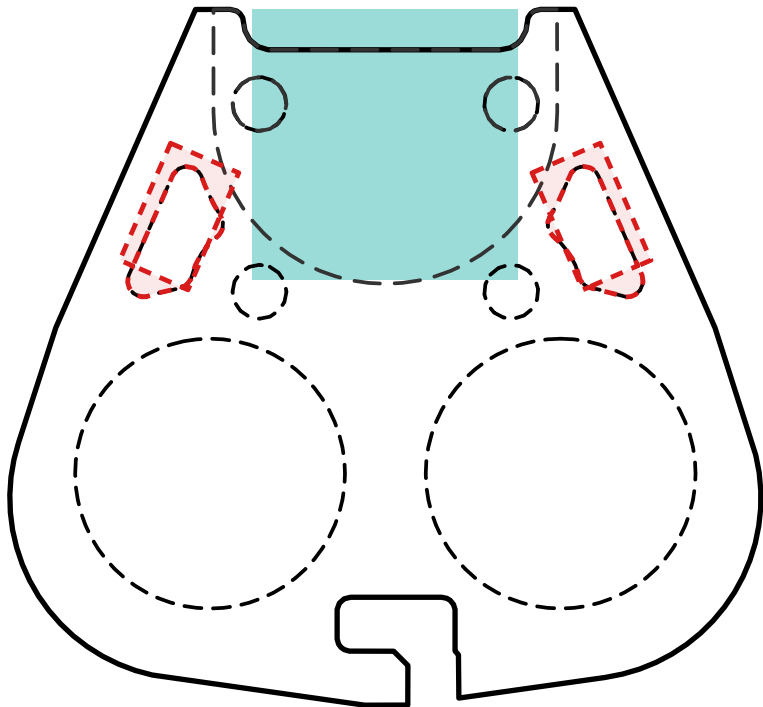
Moves: wing openings.

velcro_style

One-sided puller · Step 3 - Attachment

Strap opening style

velcro_style
Wing → Classic slot



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Swaps the curved wing openings for a pair of plain rectangular slots that lean along the body's sides.

Default	Range	Step	Unit
Wing	2 options	—	—

Options: Wing, Classic slot.

Before: Wing. After: Classic slot.

Moves: wing openings.

Can trip: WING OPENING SMALLER THAN STRAP WIDTH.

strap_width

One-sided puller · Step 3 - Attachment

Strap width

strap_width
This dial changes no shape.
Changes no shape: it only sets the width the W-14 check compares the wing opening against.

Sets the strap width the wing opening is checked against and changes no shape.

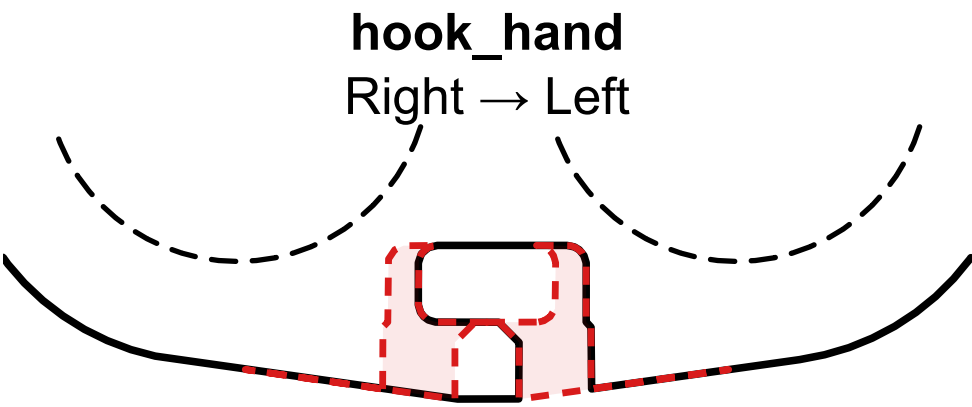
Default	Range	Step	Unit
15	10 to 25	1	mm

This dial changes no shape. Changes no shape: it only sets the width the W-14 check compares the wing opening against.
Can trip: WING OPENING SMALLER THAN STRAP WIDTH; WING WEB COLLAPSED - NO ROOM FOR STRAP.

hook_hand

One-sided puller · Step 4 - Cord Hook

Hook side



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Mirrors the cord hook at the cord end so its catch faces the other side, and nothing else moves.

Default	Range	Step	Unit
Right	2 options	—	—

Options: Right, Left.
Before: Right. After: Left.
Moves: hook.

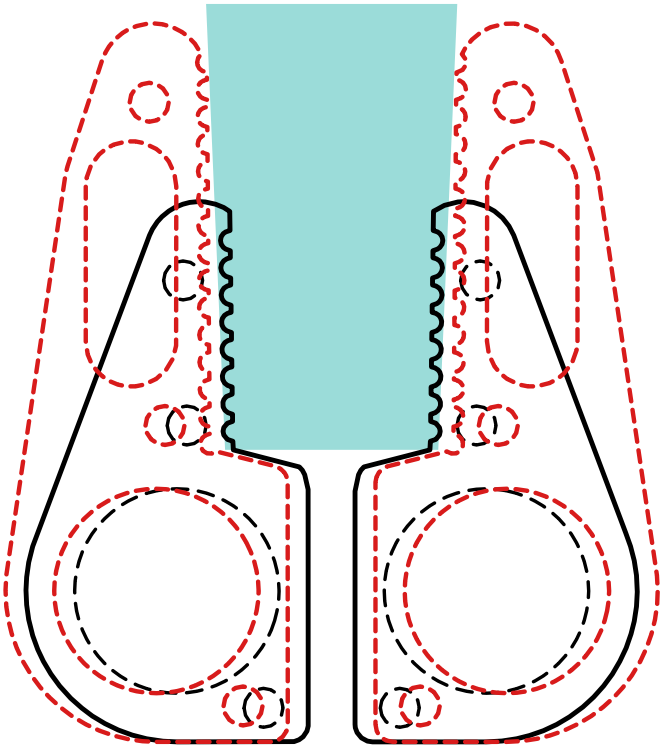
plug_preset

Two-sided puller · Step 1 - Your Plug

Plug preset

plug_preset

Measure my plug → Heavy-duty extension cord - NEMA 5-15



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Fills in the plug's length, both widths and the cord from the measured heavy-duty cord plug, so the arms, the grip gap, the cord channel and the plate length all take that plug's shape at once.

Default	Range	Step	Unit
Measure my plug	5 options	—	—

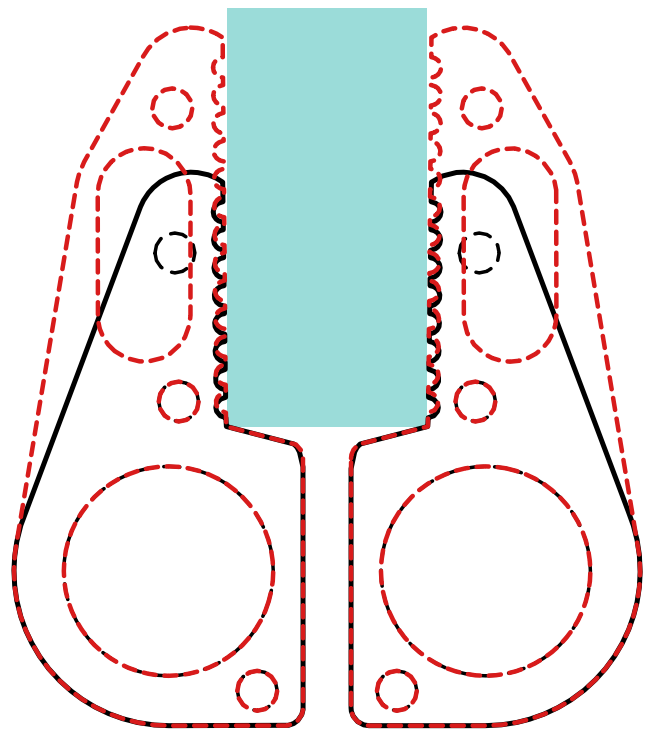
Options: Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15.
Before: Measure my plug. After: Heavy-duty extension cord - NEMA 5-15.
Moves: arms, finger lobes, plate edge, zip stations.

measure_plug_length

Two-sided puller · Step 1 - Your Plug

Plug length

measure_plug_length
25.5 → 40



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them.

Default	Range	Step	Unit
25.5	12 to 85	0.5	mm

Before: 25.5. After: 40.

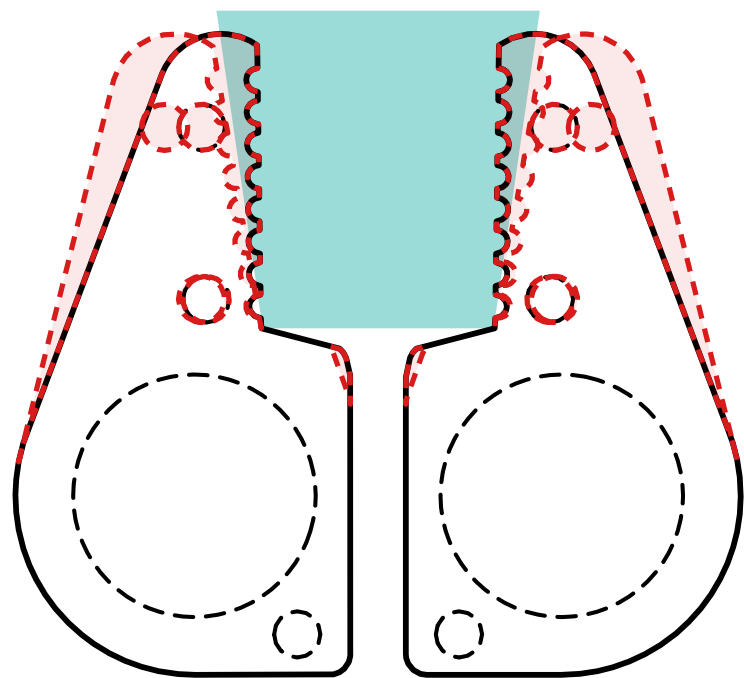
Moves: arms.

measure_plug_width_prong_end

Two-sided puller · Step 1 - Your Plug

Plug width at the prong end

measure_plug_width_prong_end
20 → 28



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Opens or closes the gap between the arms at their tips, where the plug's prong end sits.

Default	Range	Step	Unit
20	5 to 40	0.5	mm

Before: 20. After: 28.

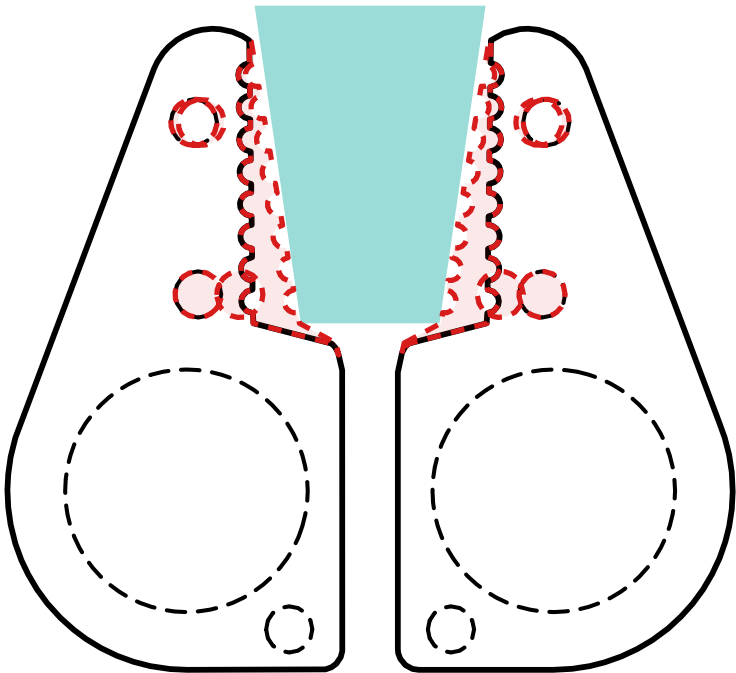
Moves: arms, cord channel, finger lobes, zip stations.

measure_plug_width_cord_end

Two-sided puller · Step 1 - Your Plug

Plug width at the cord end

measure_plug_width_cord_end
20 → 12



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Opens or closes the gap between the arms at the plug's back end, so the arms taper to match the plug.

Default	Range	Step	Unit
20	5 to 40	0.5	mm

Before: 20. After: 12.

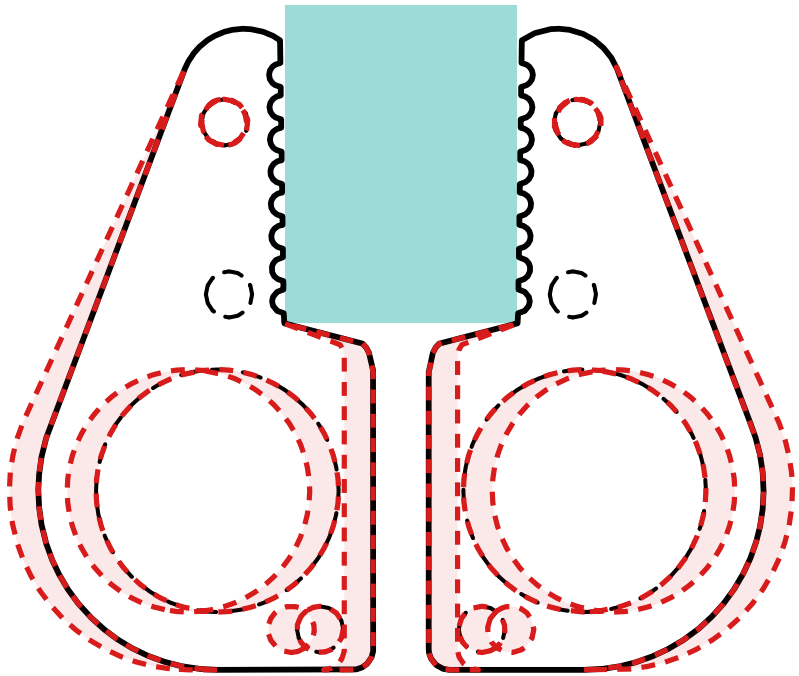
Moves: arms, teeth, zip stations.

measure_cord_thickness

Two-sided puller · Step 1 - Your Plug

Cord thickness

measure_cord_thickness
4 → 9



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens the cord channel between the finger holes, and the finger lobes move outward with it.

Default	Range	Step	Unit
4	1.5 to 12	0.5	mm

Before: 4. After: 9.

Moves: arms, cord channel, finger lobes, zip stations.

plug_sides

Two-sided puller · Step 1 - Your Plug

Plug sides

plug_sides
Rounded sides → Flat sides
(section-y: 45)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Removes the sloped cradle from the arms' gripping edges so the teeth bite straight along a flat-sided plug.

Default	Range	Step	Unit
Rounded sides	2 options	—	—

Options: Rounded sides, Flat sides.
Before: Rounded sides. After: Flat sides. Section at y = 45 mm.
Moves: arms, teeth.

show_plug_preview

Two-sided puller · Step 1 - Your Plug

Show the see-through plug

show_plug_preview

This dial changes no shape.

Changes no shape: the see-through plug is a preview aid and is never exported.

Shows or hides the see-through plug in the preview and changes nothing in the printed plates.

Default	Range	Step	Unit
true	true / false	—	—

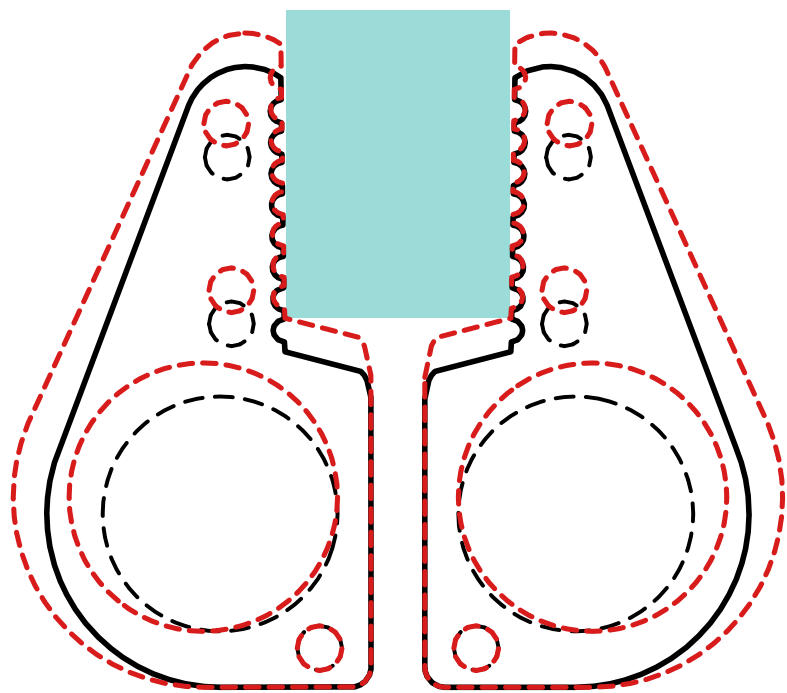
This dial changes no shape. Changes no shape: the see-through plug is a preview aid and is never exported.

size

Two-sided puller · Step 2 - Size

Hand size

size
Medium → Large



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes and their lobes, so the whole plate grows around them.

Default	Range	Step	Unit
Medium	4 options	—	—

Options: Small, Medium, Large, Measure my hand.

Before: Medium. After: Large.

Moves: arms, finger lobes, zip stations.

measure_finger_width

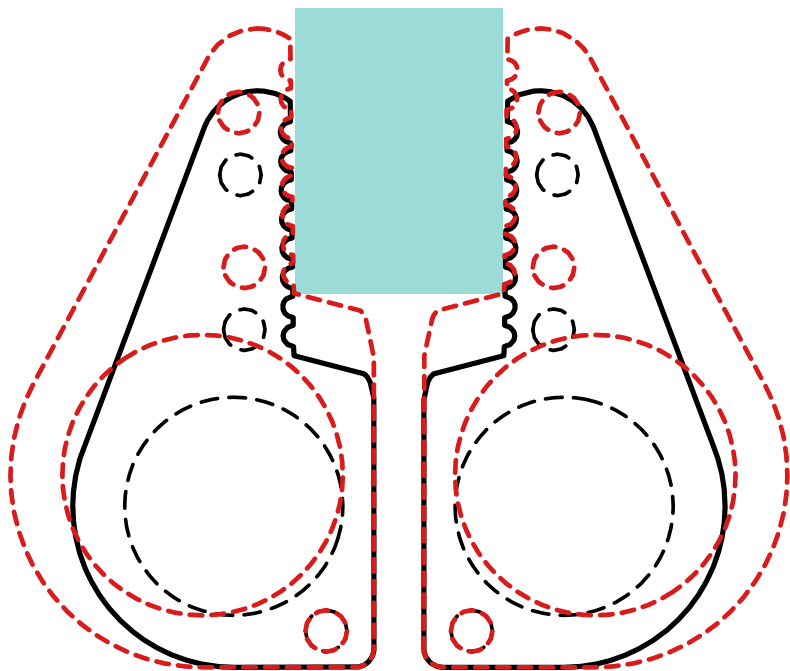
Two-sided puller · Step 2 - Size

Finger width

measure_finger_width

20 → 26

(size: Measure my hand)



black = the tool at its defaults;
teal = the plug you measured;
red dashed = what this dial moves

Widens both finger holes and their lobes.

Default	Range	Step	Unit
20	14 to 32	0.5	mm

Before: 20. After: 26. Context: size = Measure my hand.

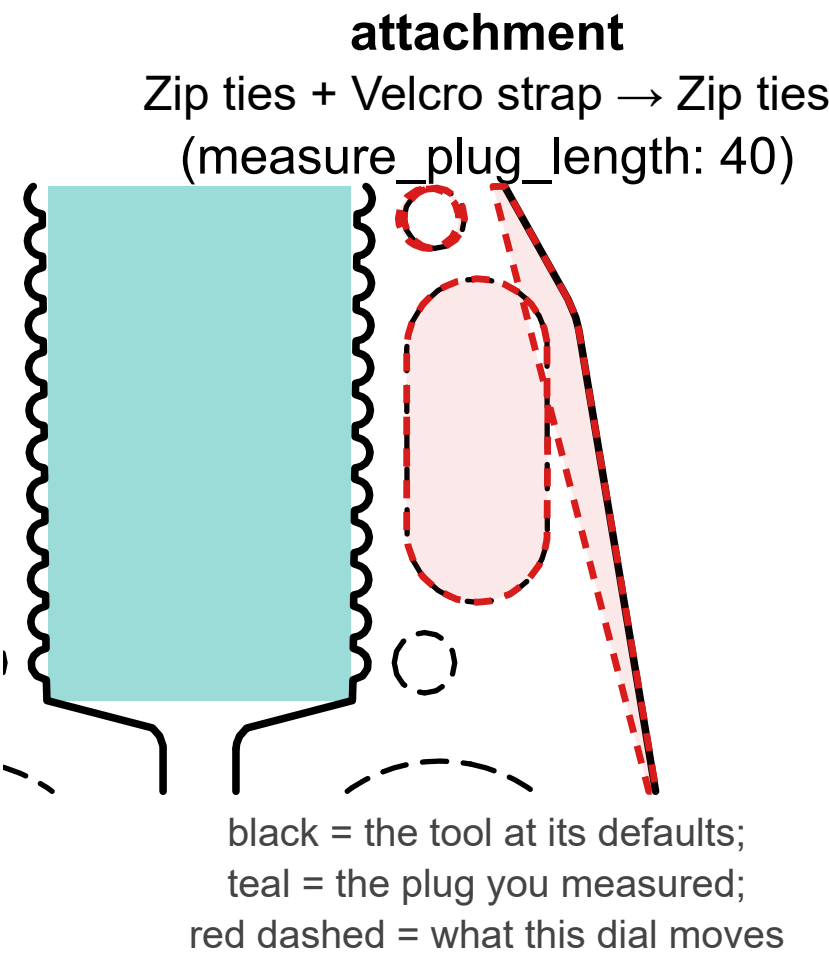
Only acts when size is Measure my hand.

Moves: arms, finger lobes, plate edge, zip stations.

attachment

Two-sided puller · Step 3 - Attachment

Attachment



Removes the strap slot from each arm, while the zip-tie stations stay because they hold the plates together.

Default	Range	Step	Unit
Zip ties + Velcro strap	2 options	—	—

Options: Zip ties + Velcro strap, Zip ties.

Before: Zip ties + Velcro strap. After: Zip ties. Context: measure_plug_length = 40.

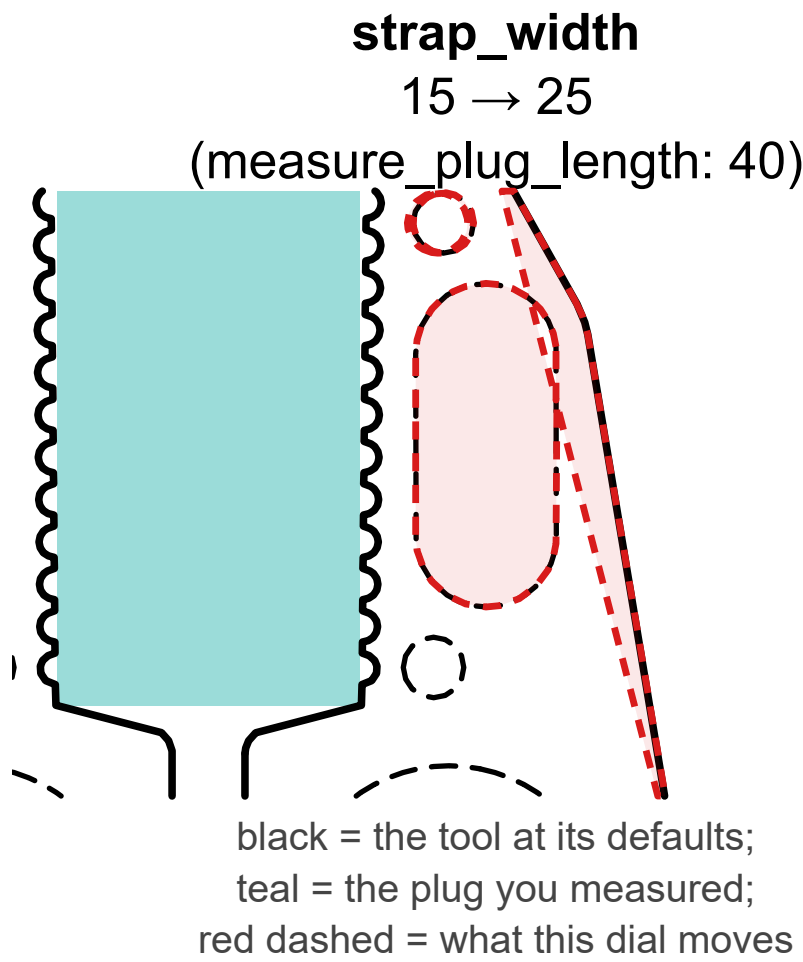
Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

strap_width

Two-sided puller · Step 3 - Attachment

Strap width



Sets the strap the arm slot must clear, and when the slot's window between the zip-tie stations is shorter than the strap plus 1.5 mm the slot is left out.

Default	Range	Step	Unit
15	10 to 25	1	mm

Before: 15. After: 25. Context: measure_plug_length = 40.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

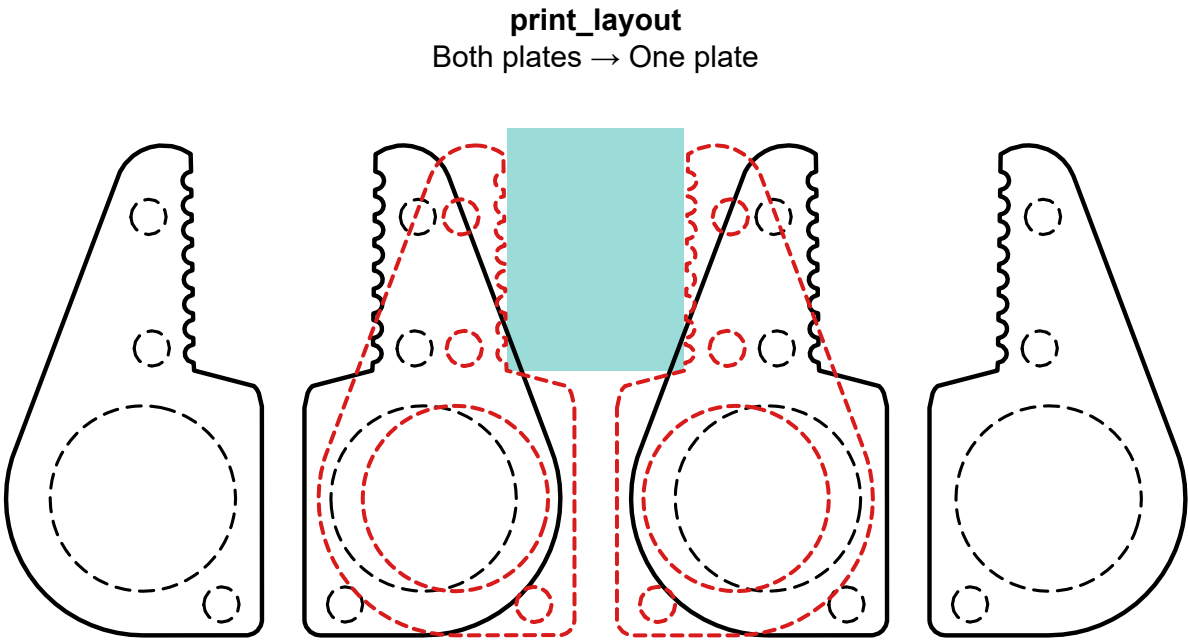
Moves: arms, strap slot, zip stations.

Can trip: STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

print_layout

Two-sided puller · Step 4 - Print Layout

Print layout



black = the tool at its defaults; teal = the plug you measured; red dashed = what this dial moves

Puts both identical plates side by side in one file, or just one plate.

Default	Range	Step	Unit
Both plates	2 options	—	—

Options: Both plates, One plate.

Before: Both plates. After: One plate.

Moves: arms, finger lobes, plate edge, zip stations.